04. ETL Data Processing Guide

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Contents

- Creating the Source In
- Amalgamating the Da
- Extra Field Calculatior
- Saved Queries
- ETL Processing
  - Stage 1
  - Stage 2
  - Stage 3
  - Stage 4
  - Stage 5
- Renaming Key Fields
- Automating the Outpu

This guide has been written to outline the steps required to Index, join, shape and flatten data using the Database Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Wizards/Database%20Data%20Source%20Wizard/), Amalgamated Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Wizards/b.%20Amalgamated%20Data%20Source%20Wizard/) and ETL Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/) to automate the creation of a customer master record, with the final output summarising customer activity across the business.

To facilitate this level of data processing, the Extract, Transform & Load (ETL) (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/) (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/)functionality is a vital component. This is due to how CXAIR (/bin/view/Technical%20Documentation/CXAIR/) processes data when Indexed on the system.

When building an Index, CXAIR (/bin/view/Technical%20Documentation/CXAIR/) handles the retrieved data row-by-row. Therefore, there is no grouping at this point and no way to relate rows to each other. For example, the same Customer ID appearing on each row is inconsequential – CXAIR (/bin/view/Technical%20Documentation/CXAIR/) simply stores the rows in the order they are retrieved.

To discover and specify these relationships, these Indexes can then be used in the Crosstab (/bin/view/Technical%20Documentation/CXAIR/User%20Guide/02.%20Reporting/2c.%20Crosstabs/) functionality where fields can be grouped at row or column level before aggregated values are derived at run-time. For example, using a date field as a column value will order the values in the report and up output accordingly.

With this in mind, the following questions pose a problem when simply viewing the raw data:

When was a customer's first transaction? What was the value?
What is the total balance value for a single customer across multiple accounts?
What is the most recent transaction value?

The answers to these questions cannot be determined without prior grouping at customer level, as each transaction is stored as a single row.

This is where the ETL (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/) functionality and its fundamentally different approach to data processing is crucial, as it is possible to loop through data based on a set of keys to group items together. Using the wizard, a number of grouped aggregat be written out to the Index that negates the requirement for report-level relationship discovery.

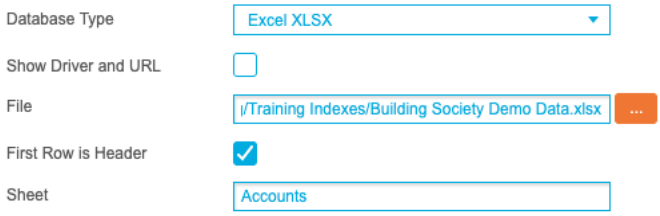
For the purpose of this guide, demonstration data has been used that contains three separate customer, account and transaction tables. For a copy of this sample dataset, p contact Connexica.

Creating the Source Indexes

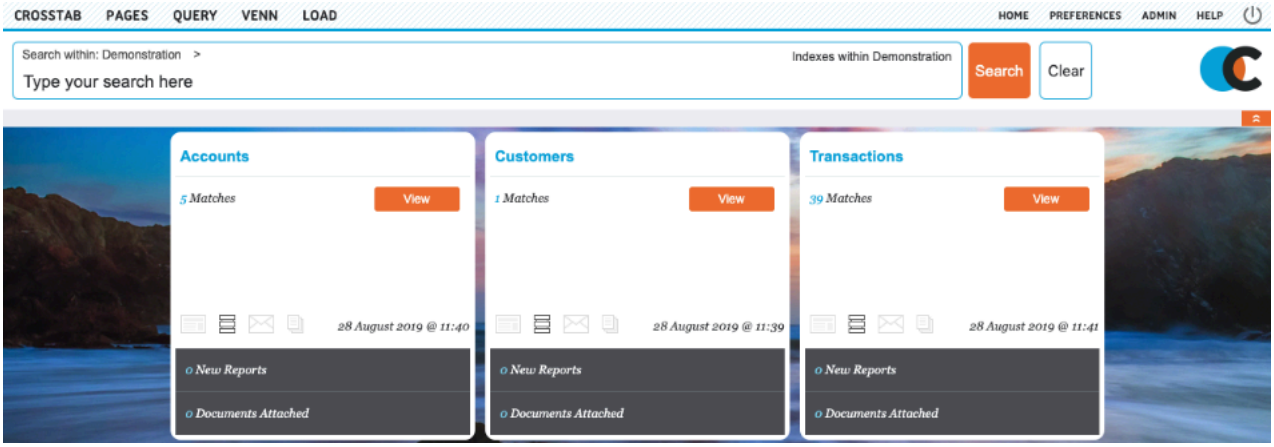
Three Indexes will be created that each serves a specific purpose, providing key information for an area of the business.

Using the Database Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Wizards/Database%20Data%20Source%20Wizard/), select Excel XLS **Database Type** and use the **File** option to select the supplied Excel file.

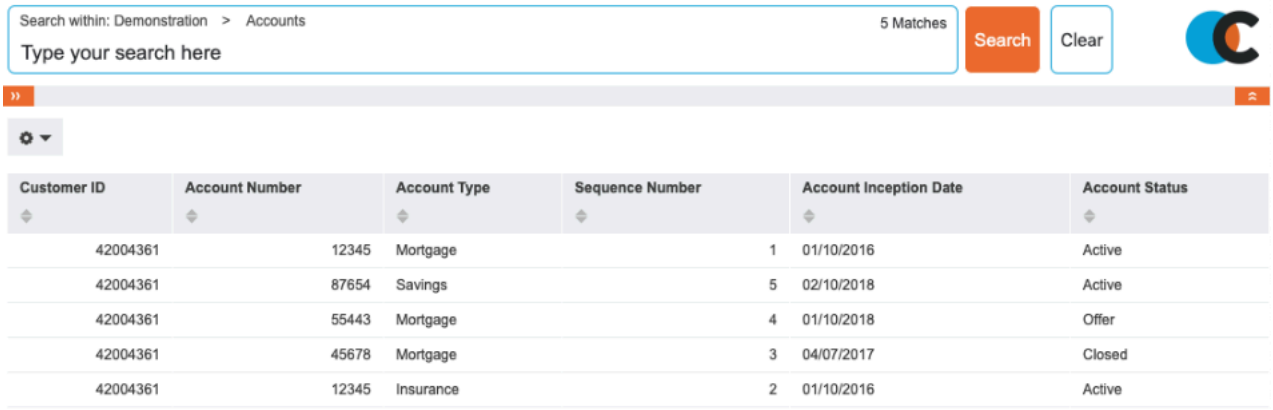
For the first Index, enter 'Accounts' into the **Sheet** textbox. For the second Index, repeat the above process and instead enter 'Customers' into the **Sheet** textbox and for the Index, enter 'Transactions' into the **Sheet** textbox:



Ensure the three Indexes are added into the same Search Engine and the Home screen should resemble the following screenshot:

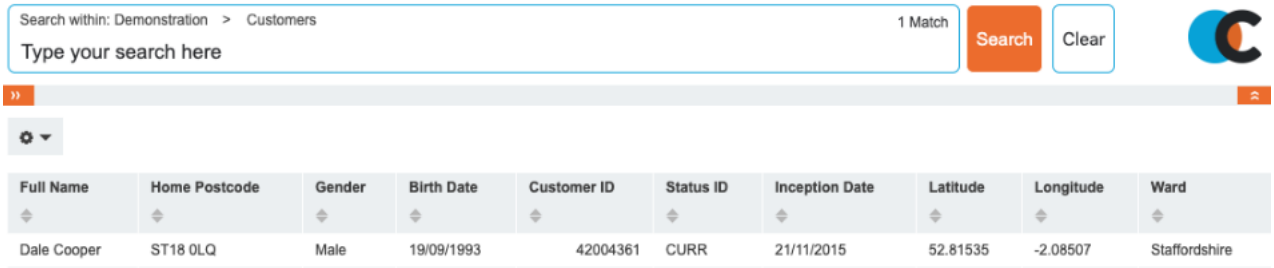


The Accounts Index contains multiple account information for a single customer:



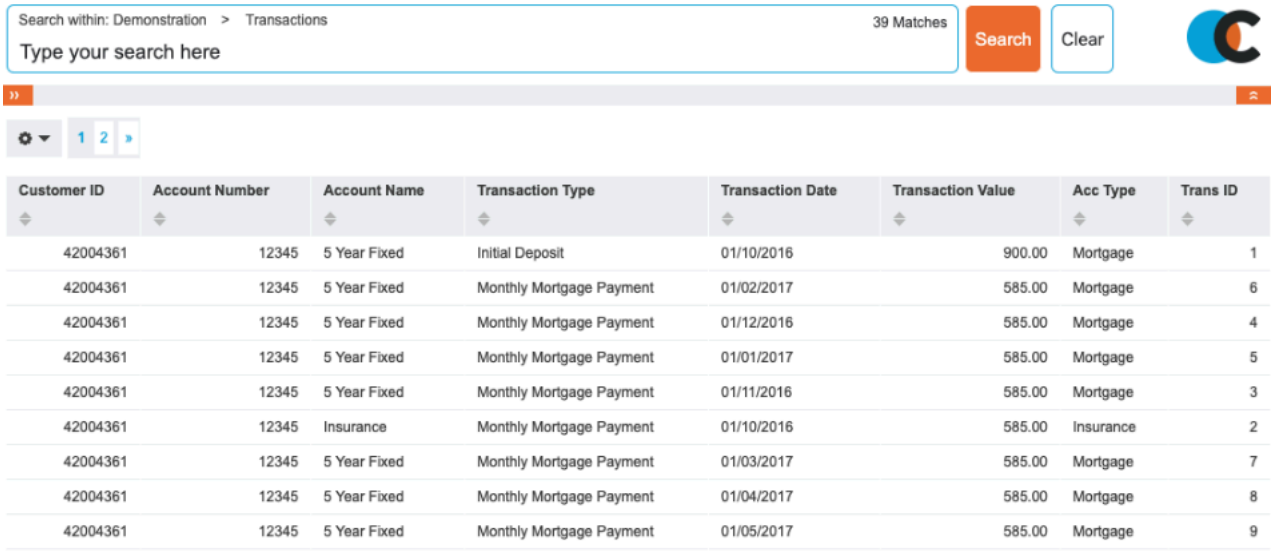
Customer ID	Account Number	Account Type	Sequence Number	Account Inception Date	Account Status
42004361	12345	Mortgage	1	01/10/2016	Active
42004361	87654	Savings	5	02/10/2018	Active
42004361	55443	Mortgage	4	01/10/2018	Offer
42004361	45678	Mortgage	3	04/07/2017	Closed
42004361	12345	Insurance	2	01/10/2016	Active

The Customers Index contains a record per customer, detailing personal information:



Full Name	Home Postcode	Gender	Birth Date	Customer ID	Status ID	Inception Date	Latitude	Longitude	Ward
Dale Cooper	ST18 0LQ	Male	19/09/1993	42004361	CURR	21/11/2015	52.81535	-2.08507	Staffordshire

The Transactions Index contains transactional-level information for a single customer across multiple accounts:



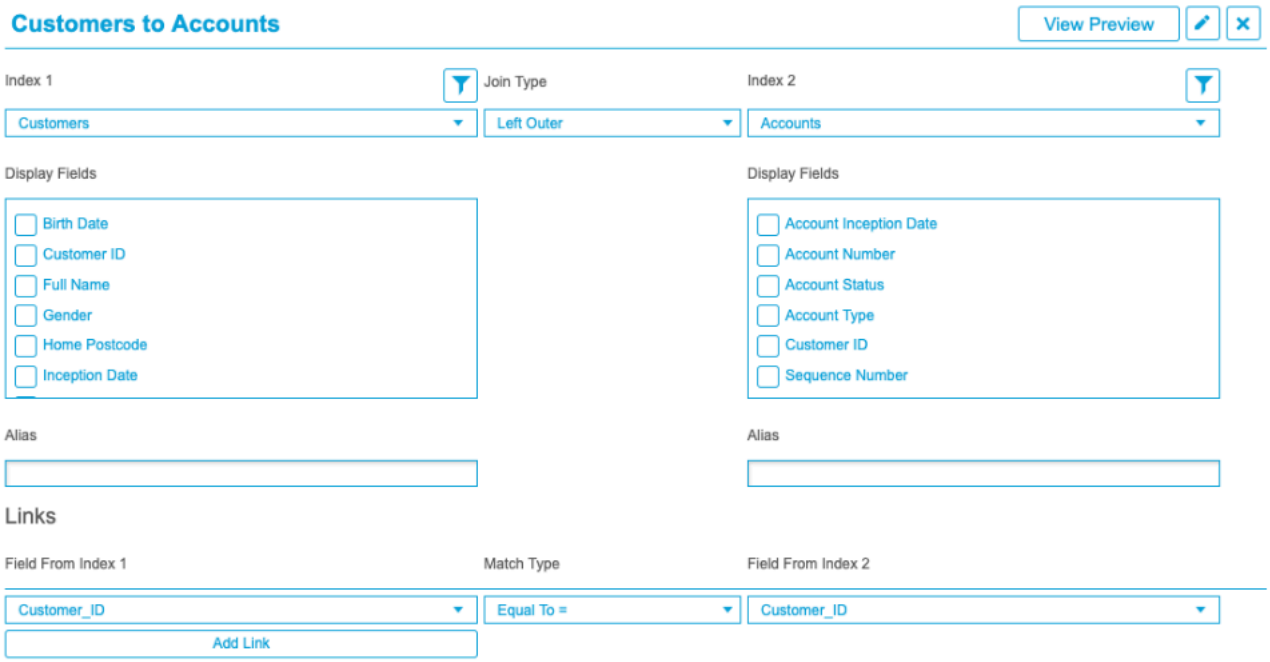
Customer ID	Account Number	Account Name	Transaction Type	Transaction Date	Transaction Value	Acc Type	Trans ID
42004361	12345	5 Year Fixed	Initial Deposit	01/10/2016	900.00	Mortgage	1
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/02/2017	585.00	Mortgage	6
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/12/2016	585.00	Mortgage	4
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/01/2017	585.00	Mortgage	5
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/11/2016	585.00	Mortgage	3
42004361	12345	Insurance	Monthly Mortgage Payment	01/10/2016	585.00	Insurance	2
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/03/2017	585.00	Mortgage	7
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/04/2017	585.00	Mortgage	8
42004361	12345	5 Year Fixed	Monthly Mortgage Payment	01/05/2017	585.00	Mortgage	9

Amalgamating the Data

With three separate Indexes now created, the Amalgamated Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Wizards/b.%20Amalgamated%20Data%20Source%20Wizard/) can be used to join the data into a s Amalgamated Index.

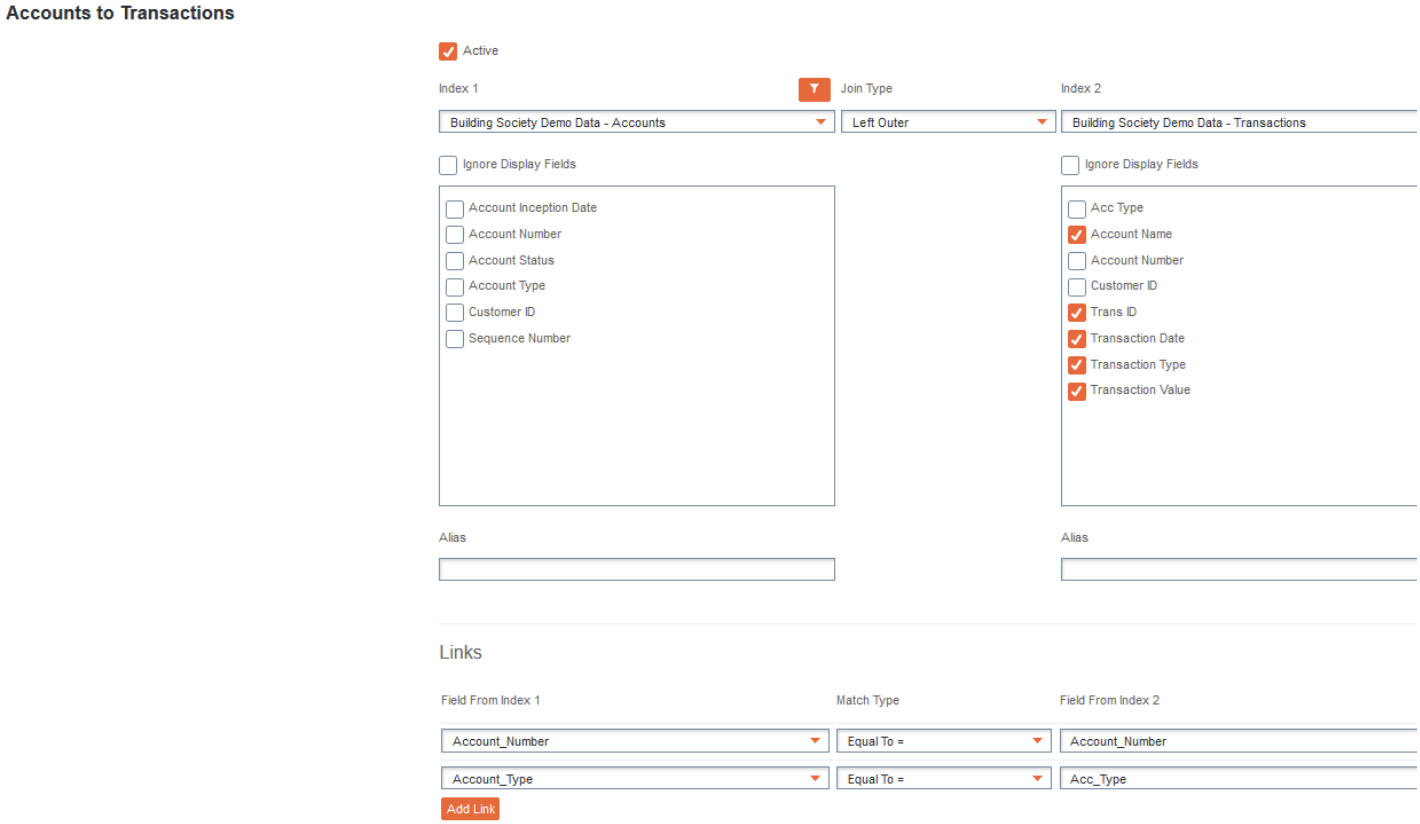
This is facilitated by two joins. First, join the Customers and Accounts Indexes using a Left Outer join and select 'Customer_ID' as the common field. This will result in accou information being added for every customer where the 'Customer_ID' field matches.

Leave the **Display Fields** checkboxes blank to include every column in the resulting output:



Next, create a second join that combines the Accounts and Transactions Indexes using a Left Outer join with 'Account_Number' used as the first common field. For this join, common field is also needed. Click **Add Link** and select 'Account_Type' and 'Acc_Type' to meet this requirement. Establishing this second condition ensures that a unique a number is not matched to the wrong type of account.

From the Accounts Index **Display Fields**, leave the checkboxes empty to bring through every column, while from the Transactions Index, select the 'Account Name', 'Trans 'Transaction Date', 'Transaction Type' and 'Transaction Value' fields to ensure duplicate fields are not in the output:



Use the **Add to Search Engine** drop-down list to specify the Search Engine the resulting Index will be added to, enable the **Build Now** option and click **Create** to complete process.

The resulting Index now contains a row for every transaction with the additional customer and account-level columns present:

Search within: Demonstration > Amalgamated Demonstration

Type your search here

40 Matches Search Clear

1 2 »

Full Name	Home Postcode	Birth Date	Customer ID	Account Name	Account Number	Account Type	Transaction Date	Transaction Type	Transaction Value	Inception Date
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/10/2016	Initial Deposit	900.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/03/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/02/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/01/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/12/2016	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/11/2016	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/04/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/05/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/06/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/07/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/09/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/10/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/11/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/12/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/02/2018	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/08/2017	Monthly Mortgage Payment	585.00	21/11/2015
Dale Cooper	ST18 0LQ	19/09/1993	42004361	5 Year Fixed	12345	Mortgage	01/08/2018	Monthly Mortgage Payment	585.00	21/11/2015

Extra Field Calculations

With an Amalgamated Index now created from the three source Indexes, the output can be further supplemented by utilising the Extra Fields (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/4.%20Manual%20Index%20Creation/b.%20Creating%20a%20Data%20Source%20Group/#Extra% functionality. Written at Data Source Group level, calculations can be added that derive additional columns at build-time.

The first extra field that can be added to the Amalgamated Index concatenates the Account Number and Account Name fields to create a composite key called Product Key. be used in later stages to denote the number of unique accounts opened by a customer and uses the following logic:

AccountNumber + '-' + AccountName

Label: Product_Key

Overwrite: ☐

Calculation: Account_Number+ '-' + Account_Name

The second extra field, Months as Customer, dynamically compares the Inception Date value for every row and compares it to the current system date. The difference is the out in months. This uses the following logic:

DateBetween(InceptionDate , TODAY , MONTHS)

Label: Months_as_Customer

Overwrite: ☐

Calculation: DateBetween(Inception_Date , TODAY , MONTHS)

Once the two calculated fields have been saved, rebuild the Index and validate the results in the Query (/bin/view/Technical%20Documentation/CXAIR/User%20Guide/02.%20Reporting/2a.%20Query/) screen.

Saved Queries

With rows from multiple accounts joined in a single Index, the following queries can be used to access key areas of interest:

+Account_Status:"Active"
+Account_Status:"Closed"
+Account_Status:"Offer"

These queries provide a cohort of data for active and closed accounts, along with any accounts that have been offered to a customer. Save these queries to utilise this dyna filtering when creating aggregations in the ETL Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/).

ETL Processing

In this example, the data processing is split into five distinct stages in order to shape the data into the final output.

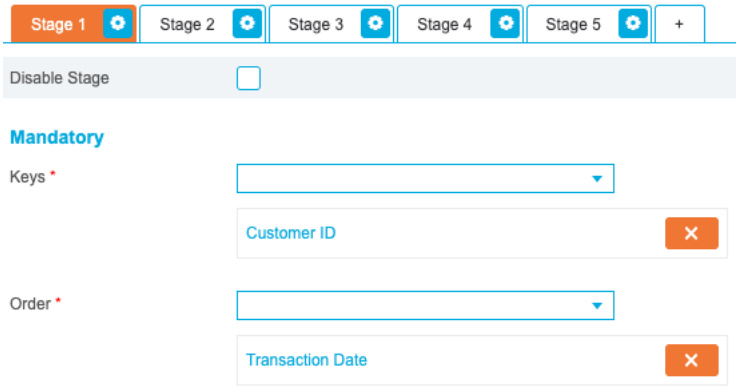
Using the ETL Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/), select the previously created Amalgamate from the **Index** drop-down list to reveal the necessary processing options. (The ETL Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/)allows additional steps to be added directly within the ETL to crea Database & Amalgamated Indexes. Clicking the '+' plus button lists these optional steps.)

Stage 1

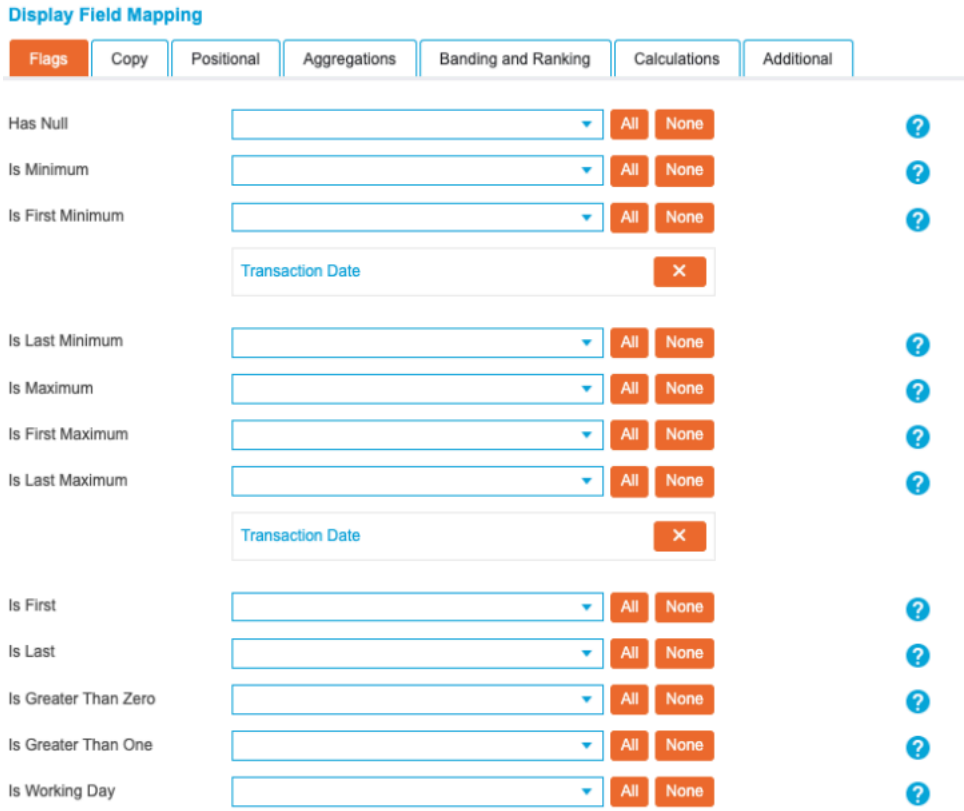
The first stage will flag the first and last transaction for each customer and will count their number of accounts, products and transactions.

When using the ETL Wizard (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Data%20Warehouse/c.%20ETL/), each stage requires a Primary Key entered in the **Key** drop-down list, and a field that will be used to order the output entered in the **Order** drop-down list.

As the output will be based at a customer-level, select 'Customer ID' from the **Key** drop-down list. Select 'Transaction Date' from the **Order** drop-down list to specify that the be ordered by the date of each transaction:

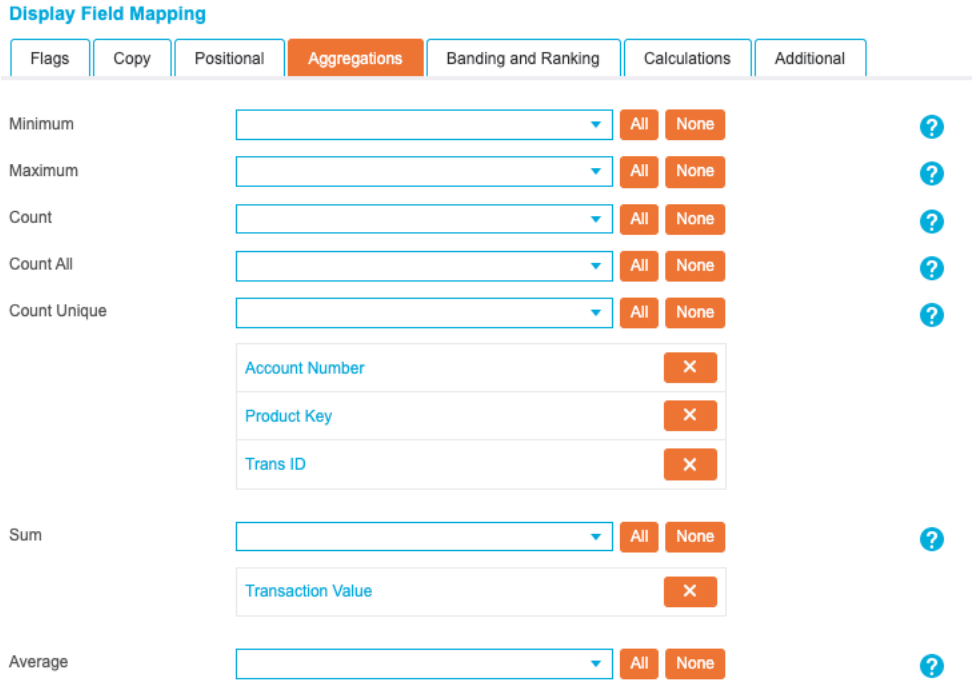


From the **Flags** tab, select the 'Transaction Date' field from the **Is First Minimum** and **Is Last Maximum** drop-down lists. This will identify the first and last transaction date customer:

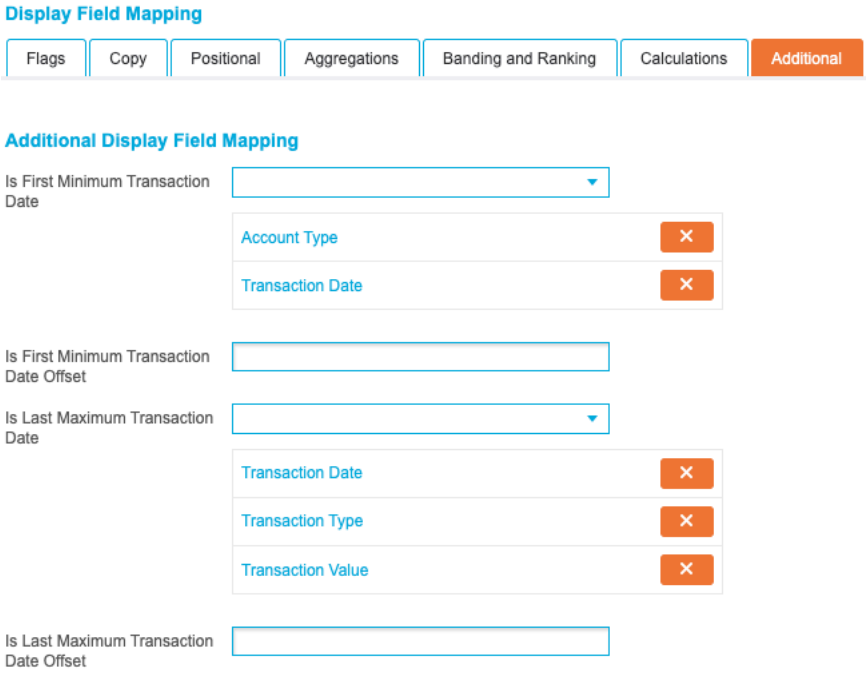


In the **Aggregations** tab, use the **Count Unique** option to count the number of accounts, products and transactions for each customer. This is achieved by selecting a key f denotes each area of interest. Select 'Account Number', 'Product Key' and 'Trans ID'.

Additionally, select the 'Transaction Value' field from the **Sum** drop-down list to sum the values from multiple rows:



As the **Is First Minimum** and **Is Last Maximum** flags have been set, the **Additional** tab provides further options to tailor the output when the flag is true. Set the 'Account T 'Transaction Date' fields for the first transaction, and the 'Transaction Date', 'Transaction Type' and 'Transaction Value' fields for the last transaction:

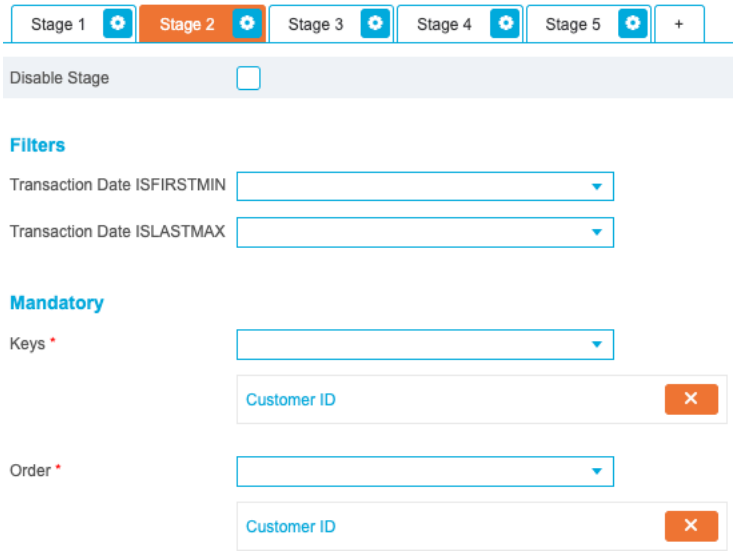


By setting these options in the **Additional** tab, each customer's first transaction is flagged and the account type is added to the row. Additionally, each customer's last transa also supplemented with the date, type of transaction and the value of the transaction.

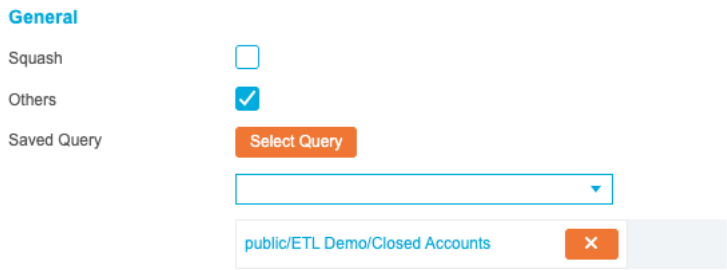
Stage 2

The next three stages all perform the same actions against different subsets of data, each pertaining to a different account type.

The second stage will count closed products per customer and does not need the data to be output in a particular order. Therefore, select 'Customer ID' for both the **Key** an drop-down lists:

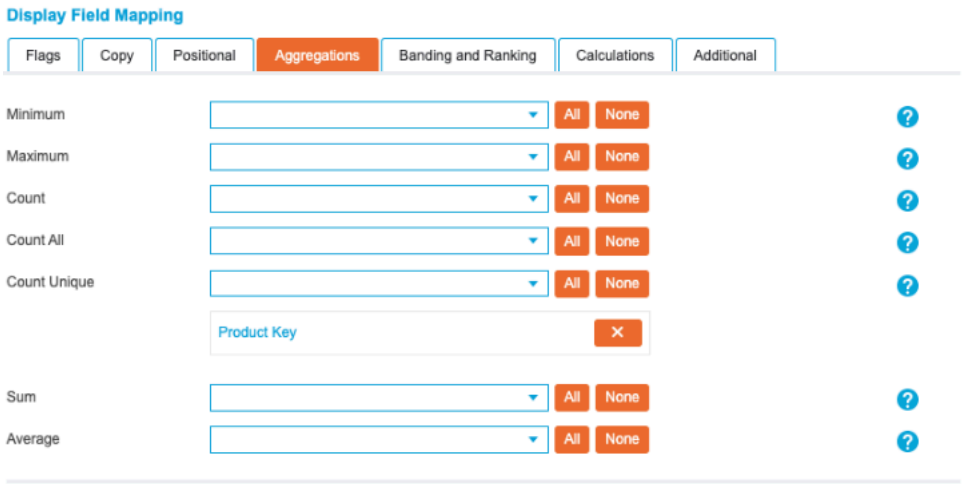


In the **General** section, select the closed accounts query saved previously and enable the **Others** option:



Using a saved query at this point will filter down this stage to rows of interest. Without the **Others** option enabled, any rows not returned by the saved query are discarded. W enabled, these results are amended to the bottom of the stage output for use in future stages. This means that stages three and four can still utilise other cohorts of data not this stage.

From the **Aggregations** tab, select the 'Product Key' field from the **Count Unique** drop-down list. Due to the added query, this will count the number of closed accounts for customer:



Stage 3

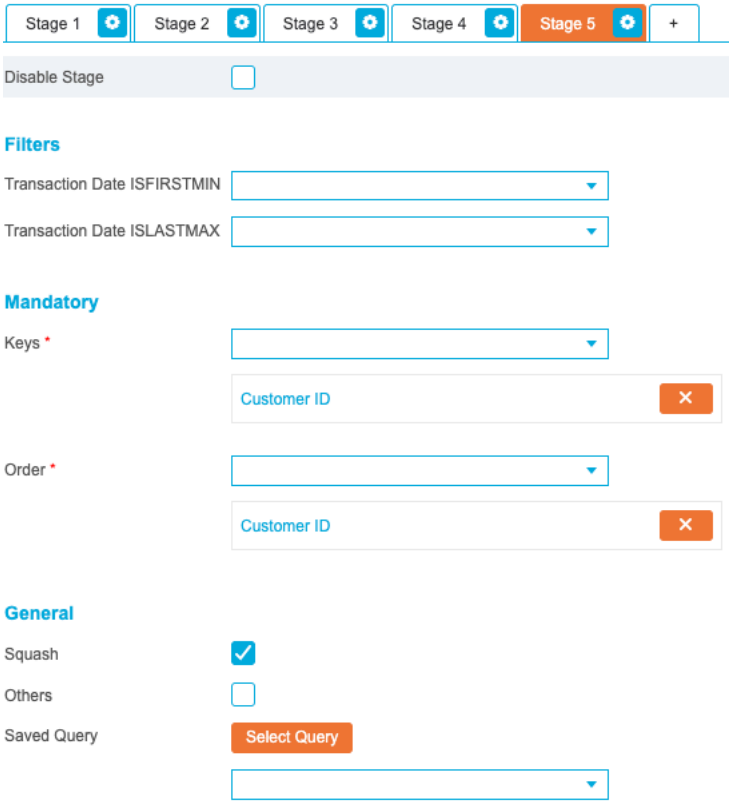
Repeat the options specified for Stage 2, instead selecting the Active Accounts query. This will count each active account per customer.

Stage 4

Repeat the options specified for Stage 2, instead selecting the Offered Accounts query. This will count each account offer per customer.

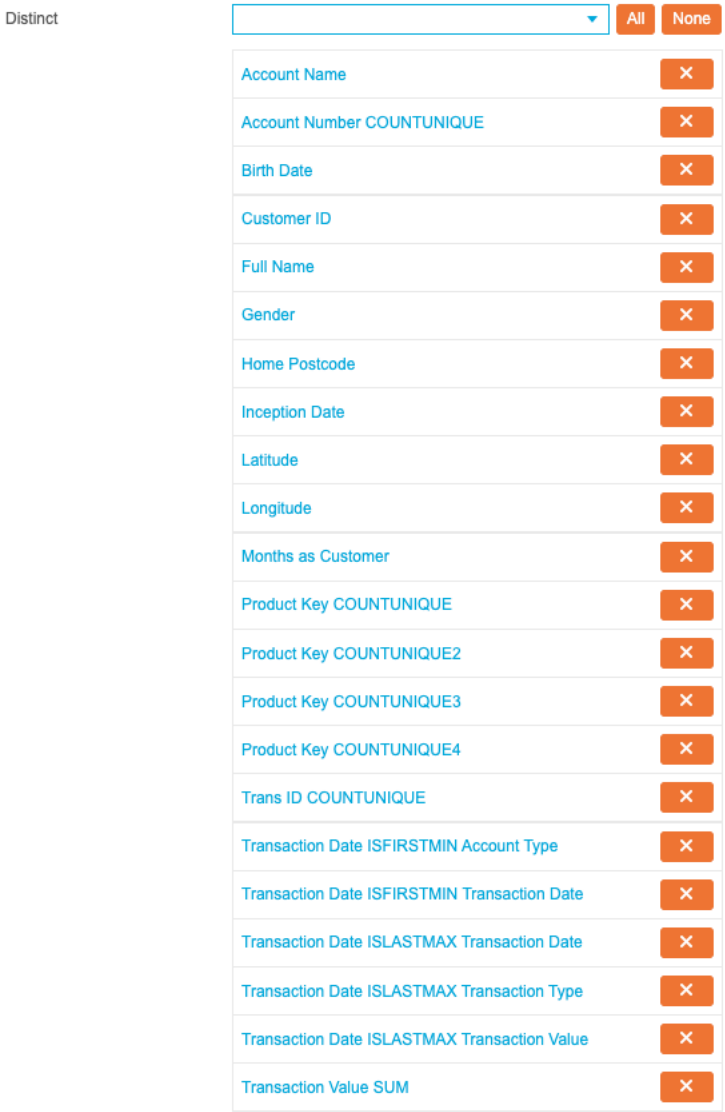
Stage 5

The final stage deduplicates records and squashes the data into a single row per customer. This is achieved by selecting 'Customer ID' in both the **Key** and **Order** drop-dow and selecting the Squash option:



When the **Squash** option has been enabled, the fields that will be present in the output now need to be specified in the **Copy** tab.

By default, every field is added to the **Copy** drop-down list. However, as the output relies on unique values, remove all entries from the **Copy** drop-down list and instead po
Distinct drop-down list with the required fields:



If preferred, these fields can instead be added to the **Unique** drop-down list at this point, which will perform the same function but order the values. As ordering the values al will not provide any meaningful insight, this is not required.

With the all five stages configured, use the **Add to Search Engine** drop-down list to specify the Search Engine the resulting Index will be added to, enable the **Build Now** o click **Create** to complete the process.

Renaming Key Fields

To enhance the end-user experience and make the fields easier to understand, navigate to the Index level of the created ETL Index and open the **Index Fields** options.

From here change the text in the **Display Field** text to better represent the field contents. For example, change 'Account_Number_COUNTUNIQUE' to 'No. of Accounts'.

For a full list of recommended field names, please see the below screenshot:

Index Field	Format	Display Field	Wrapping	Local Time Zone	Value	Add All	Remove All
Account_Name		Account_Types_Held	None Words Characters		...	Add	Remove
Account_Number_COUNTUNIQUE	#,##0	No_of_Accounts	None Words Characters		...	Add	Remove
Birth_Date	dd/MM/yyyy	Birth_Date	None Words Characters	☐	...	Add	Remove
Customer_ID	#0	Customer_ID	None Words Characters		...	Add	Remove
Full_Name		Full_Name	None Words Characters		...	Add	Remove
Gender		Gender	None Words Characters		...	Add	Remove
Home_Postcode		Home_Postcode	None Words Characters		...	Add	Remove
Inception_Date	dd/MM/yyyy	Inception_Date	None Words Characters	☐	...	Add	Remove
Latitude		Latitude	None Words Characters		...	Add	Remove
Longitude		Longitude	None Words Characters		...	Add	Remove
Months_as_Customer	#0	Months_as_Customer	None Words Characters		...	Add	Remove
Product_Key_COUNTUNIQUE	#0	No_of_Products_Purchased	None Words Characters		...	Add	Remove
Product_Key_COUNTUNIQUE 2	#0	No_of_Products_Closed	None Words Characters		...	Add	Remove
Product_Key_COUNTUNIQUE 3	#0	No_of_Products_Active	None Words Characters		...	Add	Remove
Product_Key_COUNTUNIQUE 4	#0	No_of_Products_Offered	None Words Characters		...	Add	Remove
Trans_ID_COUNTUNIQUE	#,##0	No_of_Transactions	None Words Characters		...	Add	Remove
Transaction_Date_ISFIRSTMIN_Account_Type		First_Product_Type_Opened	None Words Characters		...	Add	Remove
Transaction_Date_ISFIRSTMIN_Transaction_Date	dd/MM/yyyy	Date_of_First_Transaction	None Words Characters	☐	...	Add	Remove
Transaction_Date_ISLASTMAX_Transaction_Date	dd/MM/yyyy	Date_of_Last_Transaction	None Words Characters	☐	...	Add	Remove
Transaction_Date_ISLASTMAX_Transaction_Type		Latest_Transaction_Type	None Words Characters		...	Add	Remove
Transaction_Date_ISLASTMAX_Transaction_Value	#,##0.00	Last_Transaction_Value	None Words Characters		...	Add	Remove
Transaction_Value_SUM	#,##0.00	Total_Transaction_Value	None Words Characters		...	Add	Remove

In the Query (/bin/view/Technical%20Documentation/CXAIR/User%20Guide/02.%20Reporting/2a.%20Query/) screen the Index should now resemble the following screensh

Search within: ETL Demonstration > ETL Demo
Type your search here
1 Match
Search
Clear

Full Name	Customer ID	Gender	Birth Date	Home Postcode	Inception Date	No of Accounts	Account Types Held	Date of First Transaction
Dale Cooper	42004361	Male	19/09/1993	ST18 0LQ	21/11/2015	4	1 Year Fixed 5 Year Fixed Insurance ISA	01/10/2016

This ETL output contains a row per customer, outlining a number of key metrics to provide a view of activity across numerous areas of the business.

Automating the Output

With the data processing now complete, schedules can now be established to automate the entire process and account for new and changed records.

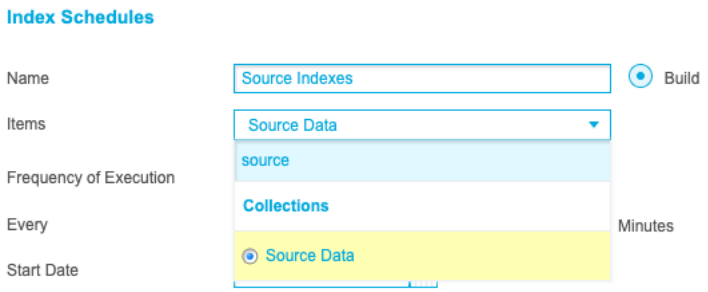
Rather than creating individual schedules for each component, this will be achieved using three Index schedules. The first will rebuild the Indexes from the source Excel she second will rebuild the subsequent Amalgamated Index and the third will rebuild the ETL Index.

First, navigate to the System Overview (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Status%20Monitoring/System%20Overview/) screen and se three source Indexes:

Data Source		Data Source Group		Index		Last Updated
Accounts	☑	Accounts	☑	Accounts	☑	30/08/2019 09:28:00
Amalgamated Demonstration	☐	Amalgamated Demonstration	☐	Amalgamated Demonstration	☐	03/09/2019 16:56:45
Customers	☑	Customers	☑	Customers	☑	30/08/2019 09:18:43
ETL Demo	☐	ETL Demo	☐	ETL Demo	☐	03/09/2019 17:00:40
Transactions	☑	Transactions	☑	Transactions	☑	30/08/2019 09:27:21

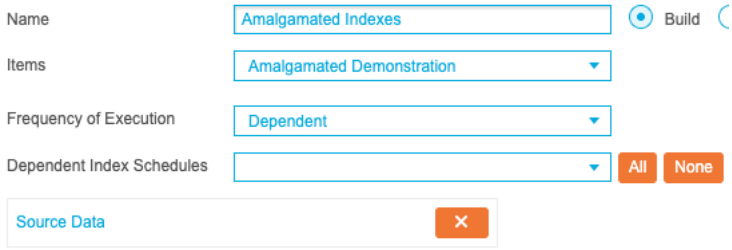
Once selected, click Collections (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Status%20Monitoring/System%20Overview/#Collections) then **Sa** name the Collection 'Source Data'. By grouping these three Indexes into one Collection, a single Index schedule can be created that will queue all contained Indexes at buil negating the requirement for many Index schedules.

Navigate to the Index Schedules (/bin/view/Technical%20Documentation/CXAIR/Administration%20Guide/Status%20Monitoring/e.%20Index%20Schedules/) screen and cli Enter 'Source Indexes' into the Name textbox and select the 'Source Data' collection from the Items drop-down list:



Use the **Frequency of Execution** drop-down list to specify how often the schedule will run and click **Create Schedule** to complete the process.

Next, create a new schedule called 'Amalgamated Indexes' and add the previously created Amalgamated Index using the **Items** drop-down list. From the **Frequency of Exe** drop-down list, select **Dependent**. This allows the schedule to only run once another schedule has completed, ensuring that the Index does not build until all of the source d been refreshed successfully. Select the 'Source Data' schedule from the **Dependent Index Schedules** drop-down list and the screen should resemble the following screens



Finally, create a third schedule titled 'ETL Indexes' and select the previously created ETL Index selected from the **Items** drop-down list and **Dependent** from the **Frequency Execution** drop-down list. This time select the 'Amalgamated Indexes' schedule. This will instruct the system to only perform the ETL processing once the Amalgamated Ind built. The screen should resemble the following screenshot:

With all three items created, the following schedules should be visible:

Source Indexes		Daily @ 00:00
Amalgamated Indexes	Amalgamated Demonstration	Dependant All(***Index Schedules:*** Source Indexes)
ETL Indexes	ETL Demo	Dependant All(***Index Schedules:*** Amalgamated Indexes)

Creating these dependent schedules will guarantee that the Indexes automatically build in the correct order, accounting for new information and changed records over time.